SKILLS ASSESSMENT MATRIX

DevSecOps Lab Book · Chapter 36 — Competency Framework

4
Roles Defined

6
Skill Areas / Role

3
Career Learning Paths

Instructor & self-assessment reference for infrastructure, security, DevOps, and application-development tracks.

Four Roles, One Framework

Each role defines required skill proficiencies and a hands-on assessment checklist.



Infrastructure Engineer

Linux, networking, IaC & Kubernetes at expert depth.

5 assessment criteria



Security Engineer

Hardening, cryptography, incident response & forensics.

5 assessment criteria



DevOps Engineer

CI/CD, orchestration, observability & automation.

5 assessment criteria



Application Developer

Secure coding, API design, testing & container basics.

5 assessment criteria

Infrastructure Engineer

Required Skills

Linux administration	Expert
Networking (TCP/IP, DNS)	Advanced
Virtualization (KVM/QEMU)	Advanced
Infrastructure as Code	Expert
Container fundamentals	Advanced
Kubernetes	Intermediate

Hands-On Assessment

- [ ] Deploy a 3-node K8s cluster in under 2 hours
- [ ] Configure network segmentation with VLANs
- [ ] Implement a disaster recovery procedure
- [ ] Scale infrastructure using IaC
- [ ] Troubleshoot etcd cluster issues

Hands-On Assessment

- Apply CIS benchmarks to systems
- Detect intrusions using IDS/IPS
- Respond to a security incident in under 1 hour
- Perform forensic analysis
- Design a secure architecture

Security Engineer

Required Skills

Linux security hardening	Expert
Cryptography fundamentals	Advanced
Network security	Advanced
Incident response	Expert
Compliance frameworks	Advanced
Threat analysis	Intermediate

DevOps Engineer

Required Skills

CI/CD pipelines	Expert
Container orchestration	Advanced
Monitoring / observability	Advanced
Automation & scripting	Expert
Release management	Advanced
Performance optimization	Intermediate

Hands-On Assessment

- Deploy an application via GitOps in under 5 minutes
- Implement auto-scaling policies
- Monitor systems with ELK or Prometheus
- Respond to performance issues
- Automate common operational tasks

Hands-On Assessment

- Write secure code that passes SAST
- Create a Docker image that passes scanning
- Deploy a Helm chart to Kubernetes
- Write security tests
- Fix critical vulnerabilities in under 24 hours

Application Developer

Required Skills

Secure coding	Advanced
Container basics	Intermediate
API design	Advanced
Testing (unit / integration)	Advanced
Kubernetes deployments	Intermediate
Security testing	Intermediate

Learning Paths

Three 8-month tracks connecting the four role competencies above.

Path 1 · Infrastructure → Security

Month 1-2	Month 3-4	Month 5-6	Month 7-8	RESULT
Linux & networking	Infrastructure as Code	Security fundamentals	Advanced security	**DevSecOps Architect**

Path 2 · Developer → DevSecOps

Month 1-2	Month 3-4	Month 5-6	Month 7-8	RESULT
Secure coding	Container security	CI/CD security	Operations	**Full-Stack DevSecOps Engineer**

Path 3 · Security → DevOps

Month 1-2	Month 3-4	Month 5-6	Month 7-8	RESULT
Security fundamentals	Linux hardening	Infrastructure	CI/CD & containers	**Security-Focused DevOps**

Using This Matrix

- Score each trainee against the six skill areas for their target role, using the three-tier proficiency scale.
- Run the hands-on assessment checklist as a practical exam — each item is a pass/fail lab task.
- Place trainees on one of the three 8-month learning paths based on their starting role.
- Revisit the matrix quarterly to track progression from Intermediate to Expert.